

# Optimization in Data Analysis: Course Projects

## *Project Proposals & Mathematical Modeling Philosophy*

### Introduction: The Philosophy of Mathematical Optimization

The core objective of these projects is not merely to write code that minimizes a function, but to understand the mathematical guarantees, architectural decisions, and modeling philosophies that make modern optimization reliable. In this course, we move away from treating algorithms as black boxes. Instead, we focus on the foundational goal of algorithm design: constructing sequences  $\{x_k\}$  such that we can mathematically guarantee  $f(x_k) - f^* \leq \epsilon$  for a selected precision  $\epsilon$ .

To achieve this, you will explore several critical paradigms:

1. **The Value of Exact Inner Solutions:** In complex algorithms, subproblems often arise. Identifying when a subproblem (like a proximal step or a 1D slice) has an exact, pen-and-paper mathematical solution drastically improves the speed and reliability of the outer algorithm.
2. **Controlling Errors and Adaptive Precision:** When using an optimization algorithm as an inner loop, computing exact solutions early on is computationally wasteful. You will learn to design dynamic stopping criteria, linking the required inner precision to the outer method's current distance from the optimum.
3. **Steering Optimization Parameters:** Parameters like step sizes, smoothing coefficients, or penalty weights should not be guessed. You will analyze convergence guarantees to determine how to dynamically update these parameters as the algorithm progresses.
4. **The Modeling Philosophy (Combinatorial & Integer Programming):** Sometimes, the algorithmic heavy lifting is done by a solver, and the true engineering challenge lies in how we formulate the problem. You will explore how replacing continuous heuristic penalties with exact discrete formulations yields drastically different computational realities.

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## 1 Estimating Sequences & Accelerated Frameworks

**Focus:** The core of accelerated methods. Students must construct and maintain the auxiliary sequences  $\phi_k(x)$  that mathematically guarantee a lower bound on the objective, ensuring  $\phi_k^* \geq f(x_k) - \epsilon$ .

### Project 1.1: Fast Gradient Method (FGM) via Estimating Sequences

**Task:** Implement FGM for smooth and strongly convex functions. You must dynamically estimate parameters (like the Lipschitz constant  $L$ ) using a line search while rigorously preserving the estimating sequence condition at every step (2 students).

**Literature:**

- Nesterov, Y. (2018). *Lectures on Convex Optimization* (2nd ed., ch. 2: Fast Gradient Methods). Springer.

### Project 1.2: Accelerated Newton's Method

**Task:** Extend estimating sequences to second-order methods, managing cubic or damped Newton steps to guarantee global accelerated convergence rates (2 students).

**Literature:**

- Nesterov, Y. (2018). *Lectures on Convex Optimization* (2nd ed., ch. 4.2). Springer.

### Project 1.3: Quasi-Newton Methods via Estimating Sequences

**Task:** Implement a Quasi-Newton scheme strictly guided by the estimating sequences framework to ensure global convergence bounds, rather than relying on heuristic updates (2 students).

**Literature:**

- Nesterov, Y. (2018). *Lectures on Convex Optimization* (2nd ed., ch. 7.4). Springer.

### Project 1.4: Acceleration via Proximal Point Dynamics

**Task:** Implement Nesterov's acceleration by analyzing it through the lens of proximal point methods, relying on exact closed-form inner steps to build the sequence (2 students).

**Literature:**

- Ahn, K., Chewi, S., et al. (2020). *Understanding Nesterov's Acceleration via Proximal Point Method*. arXiv preprint arXiv:2005.00797.

### Project 1.5: Linear Coupling (An Alternative to Estimating Sequences)

**Task:** Implement acceleration by explicitly coupling a Gradient Descent step and a Mirror Descent step. Mathematically prove how combining these geometric steps yields the accelerated rate without relying on algebraic estimating sequences (3 students).

**Literature:**

- Allen-Zhu, Z., & Orecchia, L. (2017). *Linear Coupling: An Ultimate Unification of Gradient and Mirror Descent*. 8th Innovations in Theoretical Computer Science Conference (ITCS 2017).

## 2 Higher-Order Methods & Exact Subproblems

**Focus:** Utilizing exact mathematical solutions for complex subproblems (like 1D root-finding) to accelerate non-convex or high-dimensional algorithms.

### Project 2.1: Cubic Regularization of Newton’s Method

**Task:** Implement the method and solve the critical 1D cubic-regularized quadratic subproblem exactly (using a Conjugate Gradient-like method) to make the multidimensional step computationally viable for non-convex optimization (3 students).

**Literature:**

- Nesterov, Y., & Polyak, B. T. (2006). *Cubic regularization of Newton method and its global performance*. Mathematical Programming, 108(1), 177-205.

## 3 Structural Optimization

**Focus:** Exploiting specific structural assumptions about the objective function—such as minimax forms, composite structures, or compositions of smooth and convex functions.

### Project 3.1: Minimax Smoothing for Non-Smooth Functions

**Task:** Handle non-differentiable functions by approximating them through their dual structural representation. Students must design the mathematical trigger determining when and by how much to update the smoothing parameter  $\mu$  (2 students).

**Literature:**

- Nesterov, Y. (2005). *Smooth minimization of non-smooth functions*. Mathematical Programming, 103(1), 127-152.

### Project 3.2: Gradient Method for Composite Objective Functions

**Task:** Optimize structures of the form  $f(x) = g(x) + h(x)$ , where  $g$  is smooth and  $h$  is simple but non-smooth (e.g.,  $\ell_1$  norm), using the accelerated composite gradient method (2 students).

**Literature:**

- Nesterov, Y. (2013). *Gradient methods for minimizing composite functions*. Mathematical Programming, 140(1), 125-161.

### Project 3.3: Modified Gauss-Newton for Compositions

**Task:** Solve non-linear least squares problems exploiting the compositional structure  $f(x) = g(F(x))$ , providing strict worst-case global performance guarantees rather than local heuristics (2 students).

**Literature:**

- Nesterov, Y. (2007). *Modified Gauss-Newton scheme with worst-case guarantees for global performance*. *Optimisation Methods and Software*, 22(3), 469-483.

### Project 3.4: Proximal Alternating Linearized Minimization (PALM)

**Task:** Implement the PALM algorithm to solve non-convex, non-smooth structured problems (e.g., matrix factorization or sparse PCA) modeled as  $\Psi(x, y) = f(x) + g(y) + H(x, y)$ . You must mathematically derive the exact block-coordinate proximal steps and utilize the Kurdyka-Łojasiewicz (KL) property to prove the sequence's global convergence to a critical point (3 students).

**Literature:**

- Bolte, J., Sabach, S., & Teboulle, M. (2014). *Proximal alternating linearized minimization for nonconvex and nonsmooth problems*. *Mathematical Programming*, 146(1-2), 459-494.

### Project 3.5: Inertial PALM (iPALM) for Accelerated Alternating Minimization

**Task:** Extend the alternating minimization framework by incorporating inertial (heavy-ball/momentum) steps into the proximal updates. The core mathematical challenge is to prove how the inertial parameters and step sizes must be strictly bounded and dynamically steered to guarantee global convergence in a non-convex, non-smooth landscape without breaking the descent property (3 students).

**Literature:**

- Pock, T., & Sabach, S. (2016). *Inertial proximal alternating linearized minimization (iPALM) for nonconvex and nonsmooth problems*. *SIAM Journal on Imaging Sciences*, 9(4), 1756-1787.

### Project 3.6: Sequential Joint Bilinear Surrogate Minimization for Matrix Recovery

**Task:** Diverging from the standard alternating framework of PALM, this project explores a joint optimization approach for non-convex matrix recovery problems modeled as  $L(U, V) = l_A(UV) + f(U) + g(V)$ . The task is to construct an outer-linearized bilinear surrogate,  $\langle \nabla l_A(U_k V_k), UV \rangle$ , regularized by proximal penalties. Instead of alternating, you must formulate and solve this coupled inner subproblem *jointly* via a closed-form fixed-point linear system. You will mathematically analyze the exact proximal bounds required for the joint linear system to be positive definite, and compare the theoretical convergence rate and computational per-iteration cost against standard block-coordinate descent (PALM) (3 students).

## 4 Proximal Algorithms & Approximate Inner Solutions

**Focus:** Utilizing the proximal operator and analyzing what happens when the inner proximal problem cannot be solved exactly.

### Project 4.1: Proximal Algorithms with Approximate Subproblems

**Task:** Implement a proximal gradient method where the step must be approximated. You must prove convergence assuming an approximation algorithm yielding an approximate descent bounded by a factor  $\delta$  of the guaranteed exact descent, showing the telescoped proof resulting in the scaled convergence rate (2 students).

**Literature:**

- Parikh, N., & Boyd, S. (2014). *Proximal algorithms*. Foundations and Trends in Optimization, 1(3), 127-239.

## 5 Non-Smooth & Weakly Convex Optimization

**Focus:** Algorithms that accumulate historical gradient information to stabilize steps, and extending methods to non-convex but "weakly convex" landscapes.

### Project 5.1: Primal-Dual Subgradient Methods (Dual Averaging)

**Task:** Implement dual averaging, explicitly tracking primal and dual sequences and utilizing the duality gap for a rigorous, mathematically guaranteed stopping criterion (2 students).

**Literature:**

- Nesterov, Y. (2009). *Primal-dual subgradient methods for convex problems*. Mathematical Programming, 120(1), 221-259.

### Project 5.2: Error Control in Aggregate Subgradient Methods

**Task:** Implement bundle methods, enforcing strict rules for controlling approximation error to mathematically dictate when a step is "serious" versus a "null" step (3 students).

**Literature:**

- Kiwiel, K. C. (2007). *An aggregate subgradient method for nonsmooth convex minimization*. Mathematical Programming, 111(2), 303-327.

### Project 5.3: Minimization of Weakly Convex Functions

**Task:** Optimize weakly convex functions, implementing stochastic proximal-point-like steps and theoretically tracking the convergence of the gradient of the Moreau envelope (3 students).

**Literature:**

- Davis, D., & Drusvyatskiy, D. (2019). *Stochastic model-based minimization of weakly convex functions*. SIAM Journal on Optimization, 29(1), 207-239.

### Project 5.4: Cutting plane method for SVM

**Task:** Implement cutting plane method for SVM problem with  $\ell_1$  loss function and  $\ell_2$  regularization (2 students).

**Literature:**

- V. Franc, S. Sonnenburg (2008) *Optimized cutting plane algorithm for support vector machines*. In Proceedings of the 25th international conference on Machine learning (ICML 08). Association for Computing Machinery, 2008, New York, NY, USA, 320-327

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**Literature:**

- C. Teo, et al. (2010) *Bundle methods for regularized risk minimization*. Journal of Machine Learning Research, Vol. 11, 311-365.

### Project 5.6: Interior point method for SVM

**Task:** Implement interior point method for SVM problem with  $\ell_1$  loss function and  $\ell_1$  regularization (2 students).

**Literature:**

- J. Nocedal, S. Wright (2006) *Numerical optimization*. Springer-Verlag.

## 6 Incremental & Coordinate Methods

**Focus:** Decomposing large sums or high-dimensional spaces to update parameters one component at a time, requiring dynamic parameter steering.

### Project 6.1: Incremental Gradient and Proximal Methods

**Task:** Implement incremental proximal methods for Empirical Risk Minimization. You must mathematically manage the dynamic, diminishing step-size sequences required to prevent limit cycling (3 students).

**Literature:**

- Bertsekas, D. P. (2011). *Incremental gradient, subgradient, and proximal methods for convex optimization: A survey*. Optimization for Machine Learning, 2010(1), 1-38.

### Project 6.2: Pathwise Coordinate Descent for Exact Proximal Steps

**Task:** Implement coordinate descent for Elastic Net or large-scale SVMs. The algorithmic speed must come from mathematically deriving and utilizing exact closed-form 1D updates (e.g., soft-thresholding) at each coordinate step (2 students).

**Literature:**

- Friedman, J., Hastie, T., Höfling, H., & Tibshirani, R. (2007). *Pathwise coordinate optimization*. The Annals of Applied Statistics, 1(2), 302-332.

### Project 6.3: Coordinate descent method for SVM problem

**Task:** Implement coordinate descent algorithm for binary classification problem with the loss function  $\ell_2$  and the regularization  $\ell_1$  (2 students).

**Literature:**

- K.-W. Chang, C.-J. Hsieh, and C.-J. Lin (2008) *Coordinate Descent Method for Large-scale  $L_2$ -loss Linear SVM*. *Journal of Machine Learning Research* 9.

### Project 6.4: Coordinate descent for dual SVM problem

**Task:** Implement coordinate descent method for the problem dual to  $\ell_1$  loss and  $\ell_2$  regularization SVM problem (2 students).

**Literature:**

- C-Jui Hsieh, et al, (2008) *A dual coordinate descent method for large-scale linear svm*, *Proceedings of the 25th international conference on Machine learning*, 408- 415.

### Project 6.5: Block coordinate descent method for Weapon Target Assignment problem

**Task:** Implement coordinate descent method for solving approximately WTA problem (2 students).

**Literature:**

- D. Bertsimas, A. Paskov (2025) *Solving Large-Scale Weapon Target Assignment problems in seconds using Branch-Price-And-Cut*. *Naval Research Logistics*, 1–15.

## 7 Statistical Optimization: Stopping Rules & Heavy Tails

**Focus:** In stochastic environments, geometric distance is uncertain. Stopping criteria must rely on rigorous statistical confidence.

### Project 7.1: Anytime-Valid Stopping Criteria for SGD

**Task:** Implement anytime-valid stopping rules, halting the stochastic gradient algorithm the moment mathematical statistical guarantees are met, completely avoiding fixed-epoch

training (2 students).

**Literature:**

- L. Aolaritei and M. I. Jordan (2026) *Stopping Rules for Stochastic Gradient Descent via Anytime-Valid Confidence Sequences*. arXiv: 2512.13123v5.

### **Project 7.2: Parameter Bounding in Heavy-Tailed Optimization**

**Task:** Implement robust gradient estimators in heavy-tailed noise, mathematically tying algorithmic parameters (like clipping thresholds) to theoretical high-probability bounds (3 students).

**Literature:**

- Gorbunov, E., Danilova, M., & Gasnikov, A. (2021). *High-probability bounds for non-convex stochastic optimization with heavy tails*. Advances in Neural Information Processing Systems (NeurIPS 2021).

### **Project 7.4: Stochastic gradient method inspired by a secant method**

**Task:** Implement an alternative to the ADAM method inspired by a secant method (3 students).

**Literature:**

- Gulcehre, et al. (2015) *ADASECANT: Robust Adaptive Secant Method for Stochastic Gradient*. arXiv.

### **Project 7.5: Fine tuning learning rates for faster convergence**

**Task:** Implement an algorithm that adjusts the learning rates for model parameters in a way other than using the second momentum (3 students).

**Literature:**

- L.N. Smith, et al..(2018) *Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates*. arXiv:1708.07120.

### **Project 7.6: Conjugate gradient alternative to accelerated first-order methods**

**Task:** Implement the Wolfe conjugate gradient algorithm and compare to SGD methods (3 students).

**Literature:**

- D. Zhang, S. Sen (2024) *The stochastic conjugate subgradient algorithm for kernel supporting vector machines*. arXiv:2407.21091.



## 8 Combinatorial Optimization & The Modeling Philosophy

**Focus:** Exploring how exact mathematical modeling (integer variables, submodular structures) compares to continuous heuristics in discrete spaces.

### Project 8.1: Exact Sparse Regression ( $\ell_0$ norm)

**Task:** Model and solve the exact  $\ell_0$  constrained regression problem  $\min_{\beta} \frac{1}{2} \|y - X\beta\|_2^2$  subject to  $\|\beta\|_0 \leq k$  using Mixed-Integer Programming. Consider the case when number of features is greater than the number of observations (3 students).

**Literature:**

- Bertsimas, D., King, A., & Mazumder, R. (2016). *Best subset selection via a modern optimization lens*. The Annals of Statistics, 44(2), 813-852.

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**Literature:**

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### Project 8.3: Exact vs. Heuristic Formulations in Weapon Target Allocation

**Task:** Formulate the static WTA combinatorial problem. Implement both an exact mathematical formulation and a heuristic approach to analyze trade-offs between guaranteed optimality bounds and computational feasibility (3 students).

**Literature:**

- Ahuja, R. K., Kumar, A., Jha, K. C., & Orlin, J. B. (2007). *Exact and heuristic algorithms for the weapon-target assignment problem*. Operations Research, 55(6), 1136-1146.
- Kline, A., Ahner, D., & Lunday, B. (2020). *A heuristic and metaheuristic approach to the static weapon target assignment problem*. Journal of Global Optimization, 78, 791-812.

### Project 8.4: Submodular Function Minimization/Maximization

**Task:** Formulate a discrete data analysis problem as a submodular optimization problem. Implement a greedy algorithm and analyze the exact mathematical approximation guarantees (e.g.,  $1 - 1/e$  bounds) (2 students).

**Literature:**

- Bilmes, J. A. (2022). *Submodularity in Machine Learning and Artificial Intelligence*. arXiv preprint arXiv:2202.00132.